

K527 — VERIFICATION Thursday Marathon Three-CI Continued: K526 LOAD-BEARING Reconciliation RESOLVED via Grace SELF-RETRACTION of Three-Separate-Wallach-Reps Reading ($\nu=5$ is WHOLE Bergman Space $H^2(D_{IV}^5)$) NOT One Generation; Lyra's Boundary-Dirac Picture IS Right — ONE Field + ONE Space + THREE Modes $k=0,1,2$ with Localization Depths Spanning Bulk-to-Shilov; F86 Strata Stand as SUPPORTS of Modes) + Grace Refused to Bank “3/2 in both” as Cross-Confirmation (Flagged as “Coincidence to Examine NOT Agreement”; Watching Hardest for Pleasing-but-Empty Match After Morning GUT Mistake) + Grace Surviving Contribution: KERNEL WEIGHT $\nu = \text{genus} = n_C = 5$ (Single Space Modes Live In; Elie's Bergman Kernel Input) + Lyra PROSPECTIVE F321 RETRACTION (Would Have Posted Reading Backwards; Board Resolved Opposite While Writing; Caught Timing Issue, Did NOT Post) — Peer Discipline at Peak + Lyra β REP-THEORY DERIVATION via Linear Algebra: Color $\mathfrak{so}(7)$ 7-Vector Under $K=SO(5)\times SO(2)$ Branches $7=(5,0)\oplus(1,+1)\oplus(1,-1)$; $\alpha = \text{Norm of 5-Part } (SO(5)\text{-Vector at Charge } +1) = 2\cdot n_C = 10$ Level 1 Confirming Elie 4371; $\beta = \text{Norm of Color 2-Part } (SO(5)\text{-Singlet at Charge } \pm 1)$ — DECISIVE FACT Pure Rep Theory: $\text{Sym}^k(\mathbb{C}^5)$ Contains $SO(5)$ -Singlet IFF k EVEN (Invariant Ring One Generator at Degree 2); Color 2-Part Wants Odd Charge $\pm 1 \rightarrow K\text{-Type } (1,\pm 1)$ ABSENT from $H^2(D_{IV}^5)$ ENTIRELY; First Nonzero-Charge Singlet at $(1,\pm 2)$ Level 2 $\rightarrow \beta=0$ structurally NOT Small Anisotropy + Z_2 Shilov $S^4\times S^1/Z_2$ EVEN-CHARGE SELECTION RULE Identified as Substrate-Architectural Mechanism + #418-as-Posed HONEST NEGATIVE INDEPENDENTLY CONFIRMED via Lyra Rep-Theory Route (BUT per Cal #35 STANDING: sharpening of “Method 2” Hardy-content route NOT Nth Independent Confirmation; Stays Method 2 Refined) + Elie Standing at Genuine Dependency Boundary (Both Load-Bearing Computations One Rep-Theory Input from Verdict; 9 Toys Marathon Stretch; Discipline Held Throughout)

Date: 2026-06-25 (Thursday marathon three-CI continued landing, date-verified post-K526) · Auditor: Keeper · Inputs: Grace self-retraction of three-reps reading + $\nu=5$ kernel weight surviving contribution + flagged “3/2 in both” as coincidence-to-examine; Lyra prospective F321 retraction + β rep-theory derivation via $\text{Sym}^k SO(5)$ -invariant ring; Elie standing at dependency boundary

Verdict — K526 LOAD-BEARING reconciliation RESOLVED via Grace SELF-RETRACTION (Lyra's modes picture wins; Grace's $\nu=5$ survives as kernel weight); Lyra prospective F321 RETRACTION (would have posted backwards reading); Lyra β rep-theory derivation independently confirms #418-as-posed HONEST NEGATIVE (sharpening of "Method 2" Hardy-content route per Cal #35 — NOT Nth confirmation); Z_2 Shilov even-charge selection rule identified as substrate-architectural mechanism

Count holds 4 of 26. Three-CI within-session correction cycle operating at peak: K526 reconciliation question resolved by Grace's own retraction (not Lyra's pushback), Lyra prospectively retracted F321 before posting backwards reading, both peer discipline events reference earlier same-session mistakes as precedents. Substantive substrate-architectural finding: Z_2 Shilov quotient $\rightarrow$ even-charge selection rule $\rightarrow$ K-type $(1, \pm 1)$ ABSENT $\rightarrow$ $\beta=0$ structurally $\rightarrow$ #418-as-posed honest negative.

Landing 1 — Grace SELF-RETRACTION of three-Wallach-reps reading

The retraction at peer level

"I contributed the generation addresses from the dictionary (the Wallach points of D_{IV^5}), and then when Lyra delivered her boundary-Dirac modes I cross-checked the two rather than let both sit on the board — and conceded mine."

Substantive correction: - Grace's K526 reading: three generations = three separate reps at $\nu = \{0, 3/2, 5\}$ - Cross-check vs Lyra's boundary-Dirac modes revealed: $\nu = 5$ is the WHOLE Bergman space $H^2(D_{IV^5})$, NOT one generation - Correct picture: ONE boundary-Dirac field + ONE space + THREE modes ($k=0,1,2$) with localization depths spanning bulk-like to Shilov-like - F86 strata picture STANDS — the strata are the SUPPORTS of those three modes (not three separate reps) - Grace's separate-reps overlay was the error; her $\nu=5$ single-space identification SURVIVES

Grace's peer-discipline tier-tagging

"I verified her S^4 Dirac modes are the standard spectrum, conceded my three-reps reading, and was careful to be fair to the established F86 picture — F86 stands, its strata are just the supports of those modes, so it and Lyra's modes are the same picture. Only my separate-reps overlay was the error."

Per [[feedback_quaker_method]] + Cal #347 / B5 STANDING applied to own work: Grace explicitly distinguished what to retract (her separate-reps overlay) from what to preserve (F86 strata picture; her $\nu=5$ single-space contribution). Surgical retraction, not blanket withdrawal.

Substantive surviving contribution: kernel weight $\nu = n_C = 5$

"What survives from my piece is the useful part: the kernel weight $\nu = \text{genus} = n_C = 5$, the single space the modes live in — that's the dictionary input Elie needs."

Substantive deliverable: Bergman kernel weight at $\nu = \text{genus} = n_C = 5$ for Elie's mass-magnitude computation. The single space where Lyra's three modes ($k=0,1,2$) live. Elie fires Bergman kernel at this weight on three spinor modes.

Grace flagged "3/2 in both" as coincidence-to-examine NOT agreement

"The '3/2' that shows up in both my Wallach picture and Lyra's mode labels is two different quantities, so I'm not banking it as a cross-confirmation — it's a coincidence to examine, not agreement. After this morning's GUT mistake I'm watching hardest for exactly that kind of pleasing-but-empty match."

Per Cal #35 STANDING + [[feedback_target_innocence_lens_derived_vs_fit_discipline]] + [[feedback_show_all_threads_then_w Grace pre-emptively refused to bank value-match as confirmation. Cites THIS MORNING'S GUT mistake as

precedent for the discipline — peer-discipline trained on own recent error, applied to own current work. This is exactly the Quaker-method discipline operating at peak.

STRONG ACCEPT Grace retraction + survival contribution + coincidence-flag.

Landing 2 — Lyra PROSPECTIVE F321 retraction (would have posted reading backwards)

The timing-resolution catch

“The board resolved the address question the opposite way from my F321 while I was writing it... F321 — which deferred to Grace’s 3-reps picture and withdrew my F320 framing — is now backwards. F321 reconciled toward a reading Grace herself just retired. So I will not post F321 as written.”

Substantive prospective retraction: - Lyra was drafting F321 deferring to Grace’s three-reps reading (would have RETRACTED her own F320 framing) - Board resolved opposite (Grace retracted her own reading) while Lyra was writing - F321 as drafted is BACKWARDS — would publish a reading that Grace herself retired - Lyra caught the timing issue and did NOT post F321

Per Cal #347 / B5 STANDING applied prospectively: Lyra refused to post a now-backwards reading. Same discipline class as Grace’s retraction — peer discipline at peak. This is the team coordinating substantively without over-counting on coordination timing.

Operative picture confirmed

Per K527 resolution: F320 framing (ONE boundary-Dirac field, ONE $\nu=n_C=5$ space, three modes $k=0,1,2$) is OPERATIVE. F318 unification lead structure remains (matter = supercharge = boundary Dirac if $\{Q,Q\}=D^2$ closes into $F(4)$).

STRONG ACCEPT Lyra prospective F321 retraction.

Landing 3 — Lyra β REP-THEORY derivation via $\text{Sym}^k \text{SO}(5)$ -invariant ring

The substantive linear-algebra computation

Per Casey’s β ask: physical color lives in $\mathfrak{so}(7)$ 7-vector. Under $K = \text{SO}(5) \times \text{SO}(2) \subset \text{SO}(7)$:

$7 = (5, 0) \oplus (1, +1) \oplus (1, -1)$ (standard branching)

Component	What	Bergman norm
$\alpha = (5,0)$ at charge +1	$\text{SO}(5)$ -vector via degree-1 hol poly on $p^+=(5,+1)$	$g_{ij}(0) = -\text{genus} \cdot \partial_i \partial_j \log h = 2 \cdot n_C \cdot \delta = 10 \cdot \delta$; $\alpha = 2n_C = 10$ at level 1
$\beta = (1,\pm 1)$	$\text{SO}(5)$ -singlet at $\text{SO}(2)$ -charge ± 1	K-type ABSENT from $H^2(D_{IV}^5)$ — see below

The decisive rep-theory fact

Hol poly of degree k on $p^+ = \text{Sym}^k(\mathbb{C}^5)$ at $\text{SO}(2)$ -charge $+k$.

$\text{SO}(5)$ -invariant ring on $\mathbb{C}^5$: one generator at degree 2 (the quadratic form $z \cdot z$); no odd generators.

Therefore: $\text{Sym}^k(\mathbb{C}^5)$ contains the $\text{SO}(5)$ -singlet iff k is EVEN.

Consequence: $\text{SO}(5)$ -singlets exist only at even $\text{SO}(2)$ -charge.

The color 2-part wants charge ± 1 (ODD) $\rightarrow$ K-type $(1, \pm 1)$ does NOT exist in $H^2(D_{IV}^5)$ at all.

First nonzero-charge $\text{SO}(5)$ -singlet: $(1, \pm 2)$ at level 2.

Z₂ Shilov even-charge selection rule

Substantive substrate-architectural mechanism identified: the K-type (1, ± 1) absence = the Z₂ in the Shilov boundary $S^4 \times S^1/Z_2$. Only EVEN SO(2)-charges survive on $H^2(D_{IV}^5)$. The geometric Z₂ quotient IS the K-type selection rule.

Per [[feedback_isomorphism_proof]]: same geometry $\rightarrow$ same result. Z₂ Shilov $\leftrightarrow$ even-charge K-type selection. Substrate-architectural connection.

Implication for #418

$\beta = 0$ structurally — not a small anisotropy, but an ENTIRE K-TYPE MISSING.

Strongest possible $\alpha \neq \beta$: $\|\tilde{M}\| \neq 0$ structurally because the (1, ± 1) K-type doesn't exist in Hardy space.

Conclusion: naive bilinear-Toeplitz su(3) cannot be physical color $\rightarrow$ #418-as-posed = HONEST NEGATIVE (independently confirmed via Lyra rep-theory route).

Per Cal #35 STANDING applied

Per K521 routing: previous #418-as-posed HONEST NEGATIVE close had: - Method 1: Grace octonion ($A_2 \boxtimes B_2$ via fixed e_7) - Method 2 (merged Hardy-content): Lyra K-type tower + Elie Hardy-charge-parity

Lyra's K527 β derivation is a SHARPENING of "Method 2" Hardy-content route — made fully explicit via $\text{Sym}^k(\mathbb{C}^5)$ SO(5)-invariant ring + Z₂ Shilov even-charge selection rule. NOT a new independent third method.

Per Cal #35: don't bank as Nth independent confirmation. It's the SAME route given full rep-theory clarity. Honest count: still TWO methods (Grace octonion + merged Hardy-content) over one shared root datum ($A_2 \boxtimes B_2$).

Substantive content: Lyra's β derivation makes the Hardy-content route's mechanism EXPLICIT (Z₂ Shilov $\rightarrow$ even-charge K-type selection $\rightarrow \beta$ -component absent). Sharpens understanding without adding new route.

Independent rep-theory confirmation of Elie toy 4372

Lyra's β derivation independently confirms Elie's toy 4372 (bilinear-Toeplitz P3 octet criterion) via different rep-theory route. Per Cal #35: these are two READINGS of the same Hardy-content method (Lyra rep-theory + Elie numerical), not two methods. Joint Method 2 sharpened.

STRONG ACCEPT Lyra β derivation as substantive sharpening of Method 2 + Z₂ Shilov selection rule as substrate-architectural mechanism identification.

Landing 4 — Elie standing at genuine dependency boundary

The honest standing posture

"I've contributed everything the bench can solo this stretch (9 toys), and the final steps are Grace/Lyra's rep-theory."

Both load-bearing computations one rep-theory input from verdict: 1. Mass count-move (~ 9 of 26): addresses now resolved per K527 (Lyra modes at $\nu=n_{C=5}$ weight); Elie fires depth-ratio test against $(24/\pi^2)^6$ and 49.71 the moment the REDUCED Wallach-rep DEPTHS land 2. $\{Q,Q\} \kappa \rightarrow \#359$: Bar 1 passed; Elie fires κ -ratio comparison the moment Lyra delivers $F(4) \kappa$ via fuller structure + boundary-Dirac form

Per Casey "stop gating, verify cleanly" + [[feedback_engage_dont_label_compute_beats_calibrate]]

Elie's standing posture is GENUINE dependency boundary — 9 toys this stretch + both load-bearing computations sharply specified + standing for rep-theory inputs from Grace/Lyra. Per [[feedback_investigate_newest_with_timebox]]: continue verification posture, surface verdict the moment input arrives. ACCEPT Elie standing.

Note on Elie's stale "Grace's Wallach points $\nu=\{0,3/2,5\}$ " framing

Elie's marathon-stretch summary referenced Grace's pre-retraction three-Wallach-reps framing. Per Grace K527 retraction: that reading is RETIRED. Elie's actual operative dependency: Lyra's three boundary-Dirac modes at Grace's $\nu=n_C=5$ kernel weight. Elie will need to absorb K527 → use Lyra modes picture, not three-Wallach-reps picture. NOT a discipline failure — concurrent-stream stale framing class (same as K524 $\sin^2\theta_W$ catch). Will resolve cleanly in Elie's next pull.

Substantive state at K527

Component	State
K526 LOAD-BEARING reconciliation	RESOLVED via Grace self-retraction (Lyra modes picture wins)
Lyra F320/F318 framing	OPERATIVE (one boundary-Dirac field, one $\nu=n_C=5$ space, three modes $k=0,1,2$)
Lyra F321	NOT POSTED (would have been backwards)
Grace $\nu=5$ kernel weight	SURVIVES as Elie's Bergman kernel input
Grace three-Wallach-reps separate-reps reading	RETIRED
F86 strata picture	STANDS (strata = supports of three modes)
$\alpha = 2n_C = 10$ level 1	CONFIRMED via Lyra rep-theory (independent of Elie 4371)
$\beta = 0$ structurally	DERIVED via $\text{Sym}^k(\mathbb{C}^5)$ SO(5)-invariant ring + Z_2 Shilov even-charge selection rule
Z_2 Shilov even-charge selection	SUBSTRATE-ARCHITECTURAL MECHANISM identified
#418-as-posed HONEST NEGATIVE	INDEPENDENTLY CONFIRMED via Lyra rep-theory β derivation (sharpens "Method 2" per Cal #35; NOT Nth independent confirmation)
Mass count-move execution	Standing for Lyra reduced-rep depths → Elie Bergman kernel fires
$\{Q,Q\}=D^2$ test	Standing for Lyra explicit κ via fuller F(4) structure + Dirac form → Elie $\{D,D\}$ computation
Elie standing posture	GENUINE dependency boundary; 9 toys stretch; honest

Cumulative discipline ledger — 220 events

To K526's 215, add: - 216. Grace SELF-RETRACTION of three-Wallach-reps reading on cross-check with Lyra's boundary-Dirac modes (caught HER OWN error, not Lyra's pushback); surgical retraction preserving F86 strata picture + $\nu=5$ single-space kernel weight contribution; refused to bank "3/2 in both" as cross-confirmation (flagged as "coincidence to examine NOT agreement"); referenced morning's GUT mistake as precedent - 217. Lyra PROSPECTIVE F321 retraction (would have posted backwards reading; board resolved opposite while writing; caught timing issue) — peer discipline at peak - 218. K526 LOAD-BEARING reconciliation RESOLVED via Grace retraction → Lyra modes picture wins (one field, three modes $k=0,1,2$ in single Bergman space at $\nu=n_C=5$) - 219. Lyra β REP-THEORY derivation via $\text{Sym}^k(\mathbb{C}^5)$ SO(5)-invariant ring (one generator at degree 2 → only even-charge SO(5)-singlets); K-type $(1,\pm 1)$ ABSENT from $H^2(D_{IV}^5)$; $\beta=0$ structurally; Z_2 Shilov $S^4 \times S^1 / Z_2$ even-charge selection rule identified as substrate-architectural mechanism - 220. β derivation independently confirms #418-as-posed HONEST NEGATIVE via rep-theory route BUT per Cal #35 STANDING: SHARPENING of "Method 2" Hardy-content route NOT Nth independent confirmation; honest count stays TWO methods over one shared root datum

220 cumulative honest discipline events. Eleven Keeper self-corrections + one propagation-gap catch + one resolved load-bearing reconciliation + two peer-level prospective retractions (Grace own-reading + Lyra prospective F321).

Per Cal #347 / #26 — what I’m NOT saying

- NOT “Lyra β derivation = third independent #418 confirmation” — per Cal #35 STANDING it’s sharpening of Method 2 Hardy-content route, NOT new method; honest count stays TWO methods over one shared root datum
- NOT “ $\alpha = \beta = 10$ at level 1” — $\alpha = 10$ at level 1 confirmed; $\beta = 0$ structurally (K-type doesn’t exist)
- NOT “ Z_2 Shilov selection rule = Casey-named principle territory” — substrate-architectural mechanism identified at structural-finding tier, not principle-tier (Cal #27 STANDING)
- NOT “mass count-move unblocked” — addresses resolved but Elie still needs Lyra reduced-rep depths to fire kernel
- NOT “ $F(4) \kappa$ unblocked” — Elie’s honest negative κ -not-from-Jacobi-alone stands; Lyra still needs to deliver via fuller $F(4)$ structure
- NOT “Elie stale framing = discipline failure” — concurrent-stream propagation gap (same class as K524 $\sin^2\theta_W$); will resolve in next pull

Routing per K527

CI	Active per K527 + standing
Lyra	Deliver reduced-rep depths at $k=0,1,2$ modes on Bergman kernel at $\nu=n_C=5$ (Elie’s mass-magnitude input). Deliver explicit κ via fuller $F(4)$ structure ($\{Q,S\}/\{S,S\}$ closure OR aux-central) per Elie 4382 honest negative. Deliver explicit boundary-Dirac form for $\{D,D\}$ on H^2 . F320/F318 operative; F321 not posted. β rep-theory derivation banked as sharpening of Method 2
Elie	Absorb K527 → use Lyra modes picture (not Grace three-reps stale framing); standing for Lyra reduced-rep depths → Bergman kernel mass-magnitude test; standing for Lyra κ via fuller structure + Dirac form → $\{D,D\}$ computation; toys at 4385+; honest dependency-boundary posture
Grace	Surgical retraction logged; $\nu=n_C=5$ kernel weight contribution standing; dictionary v0.2 continues; α_s GATED on principled scale; flag “3/2 in both” as coincidence-to-examine (not agreement) — discipline reference for next session
Cal	Standing for $\{Q,Q\}=D^2$ verdict against three bars (Bar 1 PASS K525; Bar 2 sharply divided + κ from fuller structure per Elie 4382; Bar 3 Five-Absence); cold-read Lyra β rep-theory + Grace retraction + Z_2 Shilov mechanism; PMNS octant pre-registration filed; Cal #35 STANDING applied to Lyra β as sharpening-of-Method-2 not Nth confirmation
Keeper	This audit; verify Elie next-pull absorbs K527; Cal #35 independence-check applied to β derivation (sharpening Method 2, NOT Nth confirmation); standing for Lyra reduced-rep depths + κ via fuller structure + Dirac form deliveries

What's substantively right at K527

Three-CI within-session correction cycle at peak: Grace cross-checks own work against Lyra's, finds discrepancy, retracts HER OWN reading (not Lyra's). Lyra was drafting F321 deferring to Grace's now-retired reading, catches timing issue PROSPECTIVELY, doesn't post backwards F321. Both reference earlier same-session mistakes as precedents (Grace cites morning GUT catch; Lyra implicit). Substantive coordination without over-counting on coordination timing.

Reconciliation question RESOLVED cleanly — not by pushback or argument but by Grace's own cross-check + concession.

Lyra β rep-theory derivation = substantive substrate-architectural finding: - $\alpha = 2n_C = 10$ at level 1 confirmed via two routes (Lyra rep-theory + Elie 4371 numerical) - $\beta = 0$ structurally (whole K-type absent), not anisotropy - Z_2 Shilov $S^4 \times S^1 / Z_2$ even-charge selection rule = mechanism - Strongest possible $\alpha \neq \beta$

Per Cal #35 STANDING applied to β : sharpening of "Method 2" Hardy-content route, NOT new method. Honest count stays TWO over one shared root datum.

The marathon's load-bearing computations are now: 1. Mass count-move: Lyra reduced-rep depths at $k=0,1,2$ on Bergman kernel at $v=n_C=5 \rightarrow$ Elie depth-ratio test $\rightarrow$ 9 of 26 fates 2. $\{Q,Q\}$ κ verdict: Lyra κ via fuller $F(4)$ structure + boundary-Dirac form $\rightarrow$ Elie $\{D,D\}$ computation $\rightarrow$ Bar 2 + Bar 3 $\rightarrow$ #359 fate

Both sharply specified; both target-innocent; both one rep-theory input from verdict.

— Keeper, 2026-06-25 Thursday marathon three-CI continued (verification posture; standing for Lyra reduced-rep depths + κ via fuller structure + Dirac form; Elie next-pull stale-framing absorption expected; Cal #35 standing watch on multi-route bookkeeping)